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Docket Nos.: 50-321

NL-25-0476

U. S. Nuclear Regulatory Commission  
ATTN: Document Control Desk  
Washington, D. C. 20555-0001

Edwin I. Hatch Nuclear Plant  
Licensee Event Report 2025-002-00

**Unit 1 Reactor Protection System Instrumentation for Turbine Control Valve Fast  
Closure Inoperable**

Ladies and Gentlemen:

In accordance with the requirements of 10 CFR 50.73(a)(2)(i)(B), 10 CFR 50.73(a)(2)(v)(A), and 10 CFR 50.73(a)(2)(vii), Southern Nuclear Operating Company hereby submits the enclosed Licensee Event Report.

This letter contains no NRC commitments. If you have any questions, please contact the Hatch Licensing Manager, Jimmy Collins, at 912.453.2342.

Respectfully submitted,

A handwritten signature in black ink, appearing to read "M. Busch", written over a horizontal line.

Matt Busch  
Vice President – Hatch

MSB/CJC

Enclosure: LER 2025-002-00

Cc:    Regional Administrator, Region II  
      NRR Project Manager – Hatch  
      Senior Resident Inspector – Hatch  
      RTYPE: CHA02.004

**Edwin I. Hatch Nuclear Plant  
Licensee Event Report 2025-002-00**

**Unit 1 Reactor Protection System Instrumentation for Turbine Control Valve  
Fast Closure Inoperable**

**Enclosure**

**LER 2025-002-00**



## LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form  
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to [Infocollects.Resource@nrc.gov](mailto:Infocollects.Resource@nrc.gov), and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

## 1. Facility Name

Edwin I. Hatch Nuclear Plant Unit 1

☒ 050  
☐ 0522. Docket Number  
003213. Page  
1 OF 2

## 4. Title

Unit 1 Reactor Protection System Instrumentation for Turbine Control Valve Fast Closure Inoperable

## 5. Event Date

## 6. LER Number

## 7. Report Date

## 8. Other Facilities Involved

| Month | Day | Year | Year | Sequential Number | Revision No. | Month | Day | Year | Facility Name | <input type="checkbox"/> 050 | Docket Number |
|-------|-----|------|------|-------------------|--------------|-------|-----|------|---------------|------------------------------|---------------|
| 10    | 22  | 2025 | 2025 | - 002 -           | 00           | 12    | 19  | 2025 | Facility Name | <input type="checkbox"/> 052 | Docket Number |

## 9. Operating Mode

1

## 10. Power Level

100

## 11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

|                                                                          |                                                    |                                                       |                                                       |                                                    |                                     |
|--------------------------------------------------------------------------|----------------------------------------------------|-------------------------------------------------------|-------------------------------------------------------|----------------------------------------------------|-------------------------------------|
| <input checked="" type="checkbox"/> 10 CFR Part 20                       | <input type="checkbox"/> 20.2203(a)(2)(vi)         | <input checked="" type="checkbox"/> 10 CFR Part 50    | <input type="checkbox"/> 50.73(a)(2)(ii)(A)           | <input type="checkbox"/> 50.73(a)(2)(viii)(A)      | <input type="checkbox"/> 73.1200(a) |
| <input type="checkbox"/> 20.2201(b)                                      | <input type="checkbox"/> 20.2203(a)(3)(i)          | <input type="checkbox"/> 50.36(c)(1)(i)(A)            | <input type="checkbox"/> 50.73(a)(2)(ii)(B)           | <input type="checkbox"/> 50.73(a)(2)(viii)(B)      | <input type="checkbox"/> 73.1200(b) |
| <input type="checkbox"/> 20.2201(d)                                      | <input type="checkbox"/> 20.2203(a)(3)(ii)         | <input type="checkbox"/> 50.36(c)(1)(ii)(A)           | <input type="checkbox"/> 50.73(a)(2)(iii)             | <input type="checkbox"/> 50.73(a)(2)(ix)(A)        | <input type="checkbox"/> 73.1200(c) |
| <input type="checkbox"/> 20.2203(a)(1)                                   | <input type="checkbox"/> 20.2203(a)(4)             | <input type="checkbox"/> 50.36(c)(2)                  | <input type="checkbox"/> 50.73(a)(2)(iv)(A)           | <input type="checkbox"/> 50.73(a)(2)(x)            | <input type="checkbox"/> 73.1200(d) |
| <input type="checkbox"/> 20.2203(a)(2)(i)                                | <input checked="" type="checkbox"/> 10 CFR Part 21 | <input type="checkbox"/> 50.46(a)(3)(ii)              | <input checked="" type="checkbox"/> 50.73(a)(2)(v)(A) | <input checked="" type="checkbox"/> 10 CFR Part 73 | <input type="checkbox"/> 73.1200(e) |
| <input type="checkbox"/> 20.2203(a)(2)(ii)                               | <input type="checkbox"/> 21.2(c)                   | <input type="checkbox"/> 50.69(g)                     | <input type="checkbox"/> 50.73(a)(2)(v)(B)            | <input type="checkbox"/> 73.77(a)(1)               | <input type="checkbox"/> 73.1200(f) |
| <input type="checkbox"/> 20.2203(a)(2)(iii)                              |                                                    | <input type="checkbox"/> 50.73(a)(2)(i)(A)            | <input type="checkbox"/> 50.73(a)(2)(v)(C)            | <input type="checkbox"/> 73.77(a)(2)(i)            | <input type="checkbox"/> 73.1200(g) |
| <input type="checkbox"/> 20.2203(a)(2)(iv)                               |                                                    | <input checked="" type="checkbox"/> 50.73(a)(2)(i)(B) | <input type="checkbox"/> 50.73(a)(2)(v)(D)            | <input type="checkbox"/> 73.77(a)(2)(ii)           | <input type="checkbox"/> 73.1200(h) |
| <input type="checkbox"/> 20.2203(a)(2)(v)                                |                                                    | <input type="checkbox"/> 50.73(a)(2)(i)(C)            | <input checked="" type="checkbox"/> 50.73(a)(2)(vii)  |                                                    |                                     |
| <input type="checkbox"/> OTHER (Specify here, in abstract, or NRC 366A). |                                                    |                                                       |                                                       |                                                    |                                     |

## 12. Licensee Contact for this LER

## Licensee Contact

Jimmv Collins. Licensina Manager

## Phone Number (Include area code)

912-453-2342

## 13. Complete One Line for each Component Failure Described in this Report

| Cause | System | Component | Manufacturer | Reportable to IRIS | Cause | System | Component | Manufacturer | Reportable to IRIS |
|-------|--------|-----------|--------------|--------------------|-------|--------|-----------|--------------|--------------------|
| E     | TG     | PS        | S079         | Y                  |       |        |           |              |                    |

## 14. Supplemental Report Expected

☒ No ☐ Yes (If yes, complete 15. Expected Submission Date)

## 15. Expected Submission Date

| Month | Day | Year |
|-------|-----|------|
|       |     |      |

## 16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

With Unit 1 at 100-percent power on 10/22/2025 at 0918 EDT, after performance of a past operability evaluation, it was determined that three Reactor Protection System (RPS) pressure switches for the Turbine Control Valve (TCV) Fast Closure trip on low Electrohydraulic Control (EHC) oil pressure were likely inoperable longer than allowed by Technical Specifications (TS). These three TCV RPS pressure switches failed TS Surveillances (SRs) on 4/5/2025 and 4/6/2025. Additionally, one of the same TCV RPS pressure switches failed a TS SR on 12/17/25. The failures of the TCV RPS pressure switches are attributed to elevated ambient temperatures which degraded EHC oil locally at the pressure switches, impeding proper depressurization of the EHC sensing lines during TCV actuation. Corrective actions included purging the EHC sensing lines, replacing the fast acting solenoid for one of the TCVs, installing temporary fluid conditioning units, increasing the TCV SR frequency, which helps prevent fluid stagnation and flushes contaminants, reducing varnish buildup and sticking.

The event is reportable per 10 CFR 50.73(a)(2)(i)(B) as a condition prohibited by technical specifications, 10 CFR 50.73(a)(2)(v)(A) for a condition that could have prevented fulfillment of a safety function, and 10 CFR 50.73(a)(2)(vii) for a common cause inoperability of independent channels.

**LICENSEE EVENT REPORT (LER)  
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form  
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

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|                                                                |                                                |                                  |                      |                                 |                      |
|----------------------------------------------------------------|------------------------------------------------|----------------------------------|----------------------|---------------------------------|----------------------|
| <b>1. FACILITY NAME</b><br>Edwin I. Hatch Nuclear Plant Unit 1 | <input checked="" type="checkbox"/> <b>050</b> | <b>2. DOCKET NUMBER</b><br>00321 | <b>3. LER NUMBER</b> |                                 |                      |
|                                                                | <input type="checkbox"/> <b>052</b>            |                                  | <b>YEAR</b><br>2025  | <b>SEQUENTIAL NUMBER</b><br>002 | <b>REV NO.</b><br>00 |

**NARRATIVE****EVENT DESCRIPTION**

With Unit 1 at 100-percent power on 10/22/2025 at 0918 EDT, after performance of a past operability evaluation, it was determined that three Reactor Protection System (RPS) (EIS Code: JC) pressure switches (EIS Code: PS) for the Turbine Control Valve (TCV) (EIS Code: FCV) Fast Closure trip on low Electrohydraulic Control (EHC) (EIS Code: TG) oil pressure were likely inoperable longer than allowed by Technical Specifications (TS). These three TCV RPS pressure switches failed TS SRs on 4/5/2025 and 4/6/2025. Additionally, one of the same RPS pressure switches failed a TS SR on 12/17/25.

**EVENT CAUSE ANALYSIS**

The failures of the TCV RPS pressure switches are attributed to elevated ambient temperatures which degraded EHC oil locally at the pressure switches. The degraded EHC oil prevented the TCV Fast Acting Solenoid (FAS) (EIS Code: SOL) from fully shutting which inhibited depressurization of the RPS pressure switch sensing line when the TCVs actuated. Because the sensing line did not depressurize when the TCVs were actuated, the pressure switches were unable to generate the RPS reactor trip signal on low EHC oil pressure as required.

**SAFETY ASSESSMENT AND REPORTABILITY**

The RPS pressure switches associated with the TCV Fast Closure trip are credited in the Updated Final Safety Analysis Report (UFSAR) Chapter 15 for initiating a reactor trip during both the Turbine Trip and Generator Load Reject events.

For the Turbine Trip event, the UFSAR credits both the TCV RPS pressure switches and the Turbine Stop Valve (TSV) (EIS Code: ISV) RPS instrumentation for initiating a reactor trip. During the period when the TCV RPS pressure switches were inoperable, the TSV instrumentation remained operable; therefore, a Turbine Trip event would have resulted in a reactor trip.

For the Generator Load Reject event, the UFSAR credits only the TCV RPS pressure switches for initiating a reactor trip. However, the UFSAR evaluates the condition when the reactor is not tripped by the TCV RPS instrumentation during a Generator Load Reject event. In such cases, a High Flux scram is credited to mitigate the event, concluding that no fuel damage would occur and that the event remains within acceptance criteria. Accordingly, there were no safety consequences as a result of the inoperability of the TCV RPS pressure switches.

Considering the failure history of the TCV RPS pressure switches, the simultaneous inoperability of multiple TCV RPS pressure switches, and since there is no other equipment in the same system as the TCV RPS pressure switches that would trip the reactor during a Generator Load Reject, this event is reportable per 10 CFR 50.73(a)(2)(i)(B) as a condition prohibited by TS and per 10 CFR 50.73(a)(2)(v)(A) as a condition that could have prevented fulfillment of a safety function. Additionally, the event is also reportable per 10 CFR 50.73(a)(2)(vii) due to degraded EHC oil creating a common cause condition that resulted in the inoperability of multiple, independent RPS channels.

**CORRECTIVE ACTIONS**

Corrective actions included purging the EHC sensing lines, replacing the fast acting solenoid for one of the TCVs, installing temporary fluid conditioning units, increasing the TCV SR frequency, which helps prevent fluid stagnation and flushes contaminants, reducing varnish buildup and sticking. Additional, planned corrective actions include replacing TCV insulation, replacing the remaining fast acting and test solenoids, and implementing a permanent EHC filtration upgrade.

**PREVIOUS SIMILAR EVENTS**

None.